OEIS A164660, final FORMULA (29 May 2016): Chebyshev numerators and floor products

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Abstract

We prove Anthony Browne's conjecture in OEIS A164660, final FORMULA (29 May 2016): for every positive integer n, the reduced numerator of $\int_0^1 T_{n+1}(x)\,dx$ equals the stated floor product and $(-1)^{n/\operatorname{rad}(n)}$. Both sides are $+1$ exactly when $4 \mid n$. The same calculation proves the preceding rational-integral conjecture and gives its denominator sequence A164661. More generally, replacing the exponent n in the product by any integer $r \geq 2$, and translating its interval by any integer, leaves the product unchanged. We give the exact root count for every $r \geq 1$ and correct two coefficient-table discrepancies in the linked sources.

1 The posed identity

Let T_m be the Chebyshev polynomial with $T_0(x) = 1$, $T_1(x) = x$ and $T_{m+1}(x) = 2xT_m(x) - T_{m-1}(x)$. Write

$$I_m = \int_0^1 T_m(x)\,dx = \sum_{j=1}^{m+1} \frac{[x^{j-1}]T_m(x)}{j}, \qquad a(m) = \operatorname{num}(I_m), \quad d(m) = \operatorname{den}(I_m), \tag{1}$$

where fractions are reduced and denominators positive. The sequences a and d are A164660 and A164661 [1, 4]; the summands are the rational triangle A164658/A164659 [2, 3]. We prove Browne's entire final FORMULA, including its two equalities and $n = 1$, rather than just its integral ingredient. The sequence was introduced by Wolfdieter Lang in 2009; Browne's assertion is dated 29 May 2016.

For $n \geq 1$, let $\operatorname{rad}(n) = \prod_{p \mid n} p$, with $\operatorname{rad}(1) = 1$. The quotient $n/\operatorname{rad}(n)$ is A003557 [7, 8]. Its interpretation as the number of nilpotent residues modulo n is classical; see Finch, Section 2 [10]. Browne already recorded the corresponding floor *sum* in A003557 on 11 May 2016. The proof below combines this arithmetic count with the Chebyshev integral, and also supplies a general-exponent version. An earlier complete proof of the posed identity is the pinned Lean declaration described in Section 5 [12]. The present note gives a self-contained exposition and the additional consequences below.

Theorem 1. *For every integer $n \geq 1$,*

$$a(n+1) = \prod_{k=1}^{n} \left(1 - 2\left(\left\lfloor \frac{k^n}{n} \right\rfloor - \left\lfloor \frac{k^n - 1}{n} \right\rfloor\right)\right) = (-1)^{n/\operatorname{rad}(n)} = \begin{cases} 1, & 4 \mid n, \\ -1, & 4 \nmid n. \end{cases} \tag{2}$$

All division inside a floor is rational division; in particular the subtraction by 1 precedes the second division.

2 The rational row sums and their denominators

Proposition 2. *We have $I_0 = 1$, $I_1 = 1/2$, and, for $m \geq 2$,*

$$I_m = \begin{cases} -1/(m^2-1), & m \text{ even}, \\ 1/(m+1), & m \equiv 1 \pmod 4, \\ -1/(m-1), & m \equiv 3 \pmod 4. \end{cases} \tag{3}$$

Consequently $a(0) = 1$ and, for $m \geq 1$, $a(m) = 1$ exactly when $m \equiv 1 \pmod 4$, and $a(m) = -1$ otherwise. Also

$$d(0) = 1, \qquad d(m) = \begin{cases} m^2-1, & m \geq 2 \text{ even}, \\ m+1, & m \equiv 1 \pmod 4, \\ m-1, & m \equiv 3 \pmod 4. \end{cases} \tag{4}$$

The odd branches include $m = 1$ in the second case.

Proof. The recurrence and the cosine addition formula give $T_m(\cos\theta) = \cos(m\theta)$, the normalization in DLMF 18.5.1 [11]. Substituting $x = \cos\theta$ and using the product-to-sum identity gives, for $m \geq 2$,

$$\begin{aligned} I_m &= \int_0^{\pi/2} \cos(m\theta)\sin\theta\, d\theta \\ &= \frac{1}{2}\left(\frac{1-\cos((m+1)\pi/2)}{m+1} - \frac{1-\cos((m-1)\pi/2)}{m-1} \right). \end{aligned}$$

For even m, both cosines vanish, giving $-1/(m^2-1)$. For $m \equiv 1 \pmod 4$ the cosines are $-1, 1$; for $m \equiv 3 \pmod 4$ they are $1, -1$. These give the other branches. The polynomials $T_0 = 1$ and $T_1 = x$ give the two initial values directly. Each displayed denominator is positive, and its numerator is 1 or -1, so all fractions are already reduced. This proves the numerator and denominator assertions. □

For odd $m = 4q+1$ or $4q+3$, respectively, the last two branches combine as

$$I_m = \frac{(-1)^{(m-1)/2}}{2\big(2\lfloor (m-1)/4 \rfloor + 1\big)}. \tag{5}$$

Together with the even branch and $I_0 = 1$, this proves the separate rational-integral conjecture preceding Browne's formula in A164660. Equation (4) gives A164661 in full.

3 Root counts and arbitrary exponents

The following count also determines exactly when the exponent may be changed without changing the count, and when it may be changed without changing its parity.

Proposition 3. *Let $n, r \geq 1$ be integers, write $n = \prod_{p \mid n} p^{e_p}$, and set*

$$D_r(n) = \prod_{p \mid n} p^{\lceil e_p/r \rceil}, \qquad C_r(n) = \frac{n}{D_r(n)}.$$

For every integer s,

$$\sum_{k=s+1}^{s+n} \left(\left\lfloor \frac{k^r}{n} \right\rfloor - \left\lfloor \frac{k^r - 1}{n} \right\rfloor \right) = C_r(n), \tag{6}$$

$$\prod_{k=s+1}^{s+n} \left(1 - 2 \left(\left\lfloor \frac{k^r}{n} \right\rfloor - \left\lfloor \frac{k^r - 1}{n} \right\rfloor \right) \right) = (-1)^{C_r(n)}. \tag{7}$$

For $n > 1$, the equality $C_r(n) = n/\operatorname{rad}(n)$ holds exactly when $r \geq \max_{p \mid n} e_p$; for $n = 1$ it holds for every $r \geq 1$. For every $r \geq 2$, the product in (7) equals $a(n+1)$, independently of s.

Proof. For any integer t, Euclidean division $t = qn + u$, $0 \leq u < n$, gives

$$\lfloor t/n \rfloor - \lfloor (t-1)/n \rfloor = \begin{cases} 1, & n \mid t, \\ 0, & n \nmid t. \end{cases} \tag{8}$$

Indeed, at $u = 0$ the floors are $q, q-1$, and otherwise both are q. This also proves the identity for negative t and for zero. For nonzero k, the prime-power conditions give

$$n \mid k^r \iff r v_p(k) \geq e_p \ (p \mid n) \iff D_r(n) \mid k;$$

the same equivalence holds at $k = 0$. Because $D_r(n) \mid n$, every interval of n consecutive integers contains exactly $n/D_r(n)$ multiples of $D_r(n)$. Equivalently its count is $\lfloor (s+n)/D_r(n) \rfloor - \lfloor s/D_r(n) \rfloor = n/D_r(n)$. Now (8) proves both the sum and the product. For $n > 1$, $D_r(n) = \operatorname{rad}(n)$ exactly when every $\lceil e_p/r \rceil = 1$, which is the stated sharp threshold.

It remains to determine the sign. If n is odd, $C_r(n)$ is odd. Otherwise let $t = v_2(n) \geq 1$. Then

$$v_2(C_r(n)) = t - \lceil t/r \rceil.$$

For $r \geq 2$, this is zero at $t = 1$ and positive at every $t \geq 2$, since $\lceil t/r \rceil \leq \lceil t/2 \rceil \leq t - 1$. Thus $C_r(n)$ is even exactly when $4 \mid n$. Proposition 2 identifies this sign with $a(n+1)$. □

Proof of Theorem 1. If $n \geq 2$, take $r = n$ and $s = 0$ in Proposition 3. For every prime-power divisor $p^{e_p} \mid n$, $e_p \leq 2^{e_p} \leq p^{e_p} \leq n$, so $D_n(n) = \operatorname{rad}(n)$. The product is therefore $(-1)^{n/\operatorname{rad}(n)}$, with the stated sign and numerator. At $n = 1$ the single factor is $1 - 2(\lfloor 1 \rfloor - \lfloor 0 \rfloor) = -1$, while $a(2) = -1$ and $1/\operatorname{rad}(1) = 1$. □

Taking $r = n$, $s = 0$ in (6) recovers Browne's earlier floor sum in A003557 and its reciprocal radical formula in A007947. For $r = 1$, $D_1(n) = n$, so the product is always -1; the uniform restriction $r \geq 2$ in the numerator identity is sharp, as $n = 4$ shows. The count itself need not be constant for $r \geq 2$: at $n = 16$ it is $C_2(16) = 4$ and $C_4(16) = 8$, while both products are $+1$. The familiar prime-power counts for $r = 2, 3, 4$ are respectively A000188, A000189 and A000190 [13]; Proposition 3 recovers those prior formulas. It also sharpens Bottomley's COMMENT in A003557 (18 September 2001): $n < 2^{r+1}$ implies $e_p \leq \log_2 n < r + 1$, hence $e_p \leq r$ and $C_r(n) = n/\operatorname{rad}(n)$. The exact threshold is the maximum prime exponent, not this sufficient bound. The square-indicator identity conjectured by Yanev in A000188 (4 July 2021) already has Fried's proof (2025, Theorem 1) [14]: writing $n = u^2 d$ with d squarefree, nk is a square exactly when $k = dv^2$, and $1 \leq k \leq n$ is equivalent to $1 \leq v \leq u$. There are $u = C_2(n)$ such integers. This is a prior result, not an additional open question.

4 Coefficient corrections and source coverage

Put $c_{m,j} = [x^j]T_m(x)$ and let $b_{m,j}$ be the coefficient of x^j in the zero-constant primitive. Coefficientwise integration gives

$$b_{m,j} = c_{m,j-1}/j \qquad (1 \le j \le m+1). \tag{9}$$

This is the defining link between A053120 and A164658/A164659 [9, 2, 3]. The column multiplier printed as $j/2^{j-2}$ in the COMMENT of A164658 must instead be $j/2^{j-1}$ to obtain $c_{m,j-1}/2^{j-1}$. At $(m, j) = (0, 1)$ the printed multiplier gives 2 rather than 1; equation (9) proves the corrected identity for every column. Also, Lang's linked table under A164660 prints 56/3 in rational row $n = 7$, displayed column $m = 5$, corresponding to $b_{7,6}$. The recurrence gives $T_7(x) = 64x^7 - 112x^5 + 56x^3 - 7x$, so $b_{7,6} = -112/6 = -56/3$. The table's numerator -56 and row sum $-1/6$ agree with this correction.

Source item	Coverage
A164660, final FORMULA, 29 May 2016	Proved: Theorem 1, including both equalities and $n = 1$.
A164660, preceding rational conjecture	Proved: Proposition 2 and (5).
A164661, denominator definition	Proved: the complete formula (4).
A003557, Browne floor sum, 11 May 2016	Prior formula recovered by (6) with $r = n$, $s = 0$.
A007947, Browne radical formula, 17 June 2016	Prior floor-sum equality recovered by taking its reciprocal.
A003557, Bottomley COMMENT; A000188–A000190	Prior root-count formulas recovered above; the sufficient bound is sharpened. A000188's Yanev identity has Fried's prior proof.
A164658, column COMMENT; A164660 linked table, row 7	Refuted and corrected by (9) and the displayed T_7.
A164658/A164659, coefficient definitions; A053120 recurrence	Matched by (9) and the defining recurrence for T_m.
A164662/A164663, separate numerator/denominator row sums [5, 6]	Not covered by a new closed formula; their defining finite sums are checked for the published prefixes by the accompanying program.

The last two sequences sum reduced coefficient numerators or denominators separately, not the rational coefficients, so they are distinct from I_m.

5 Formal and finite verification scope

At upstream revision `122c987d276498783de7d873378a28b27ced8761`, the Lean declaration

`D5.S3.ArithSums.BrowneA164660ChebyshevFloorProduct.result`

proves the two source equalities of Theorem 1 for every positive natural n. Its `rowSum` is exactly the finite coefficient sum in (1); `a` is its reduced signed rational numerator; `floorProduct` uses the literal rational floors. The consumed lemma `row_sum_evaluation` gives the same rational integral branches for $m = n + 1 \ge 2$. The arbitrary-exponent and translated-interval assertions of Proposition 3, the coefficient corrections, and the initial integrals I_0, I_1 have the written proofs given here. The accompanying `verify.py` uses exact integers and rational arithmetic. Its output states the finite ranges checked; those finite checks complement, rather than replace, the all-index proofs.

References

[1] W. Lang and A. Browne (formula contributor), OEIS A164660, revision 19 (2 October 2023), especially final FORMULA (29 May 2016), preceding FORMULA, and linked coefficient table. https://oeis.org/A164660.

[2] W. Lang, OEIS A164658, revision 21 (8 April 2025), FORMULA, COMMENT, and linked table. https://oeis.org/A164658.

[3] W. Lang, OEIS A164659, revision 10 (6 October 2016), FORMULA. https://oeis.org/A164659.

[4] W. Lang, OEIS A164661, revision 9 (6 March 2021), FORMULA. https://oeis.org/A164661.

[5] W. Lang, OEIS A164662, revision 8 (6 March 2021), FORMULA. https://oeis.org/A164662.

[6] W. Lang, OEIS A164663, revision 8 (6 March 2021), FORMULA. https://oeis.org/A164663.

[7] OEIS A003557, revision 172 (1 October 2026), definition and A. Browne's floor-sum FORMULA (11 May 2016). https://oeis.org/A003557.

[8] OEIS A007947, revision 324 (21 October 2026), definition and A. Browne's floor-sum FORMULA (17 June 2016). https://oeis.org/A007947.

[9] W. Lang, OEIS A053120, revision 134 (4 October 2026), coefficient recurrence. https://oeis.org/A053120.

[10] S. Finch, *Idempotents and Nilpotents Modulo n*, arXiv:math/0605019v2 (18 November 2017), Section 2. https://arxiv.org/abs/math/0605019v2.

[11] NIST Digital Library of Mathematical Functions, Section 18.5, equation 18.5.1, Chebyshev trigonometric representation. https://dlmf.nist.gov/18.5.E1.

[12] Trureturing contributors, *BrowneA164660ChebyshevFloorProduct.lean*, revision `122c987d276498783de7d873378a28b27ced8761`, declaration `result`; linked in Section 5.

[13] OEIS A000188, revision 183 (11 July 2026); A000189, revision 64 (11 July 2026); A000190, revision 41 (11 July 2026), NAME and prime-power FORMULA. https://oeis.org/A000188, https://oeis.org/A000189, https://oeis.org/A000190.

[14] S. Fried, *Proofs of some conjectures of Yanev*, 2025, Theorem 1; linked under A000188 (1 December 2025). https://oeis.org/A000188/a000188.pdf.

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